

Introduction to JavaScript

JavaScript is a programming language used to make websites interactive. It powers everything from Instagram likes to YouTube controls — transforming static pages into dynamic experiences.

Before JavaScript

- Static pages only
- No interaction
- Only text and images

After JavaScript

- Interactive & dynamic
- Real-time updates
- Rich user experiences

Websites

Instagram likes, form validation, dark mode

Mobile Apps

Cross-platform interactive apps

Games

Browser-based game logic

Backend & AI

Servers, APIs, and AI tools



Teaching Tip: Ask students — "What happens when you click the Like button on Instagram?" Then explain: JavaScript handles that interaction.

History of JavaScript

JavaScript was created by **Brendan Eich** in just **10 days** — a remarkable feat that launched the most popular web language in the world.

- 1** — **1995**
JavaScript created by Brendan Eich
- 2** — **1996**
Adopted by Netscape browser
- 3** — **Today**
Most popular web language worldwide

HTML

Structure of the page

CSS

Design and styling

JavaScript

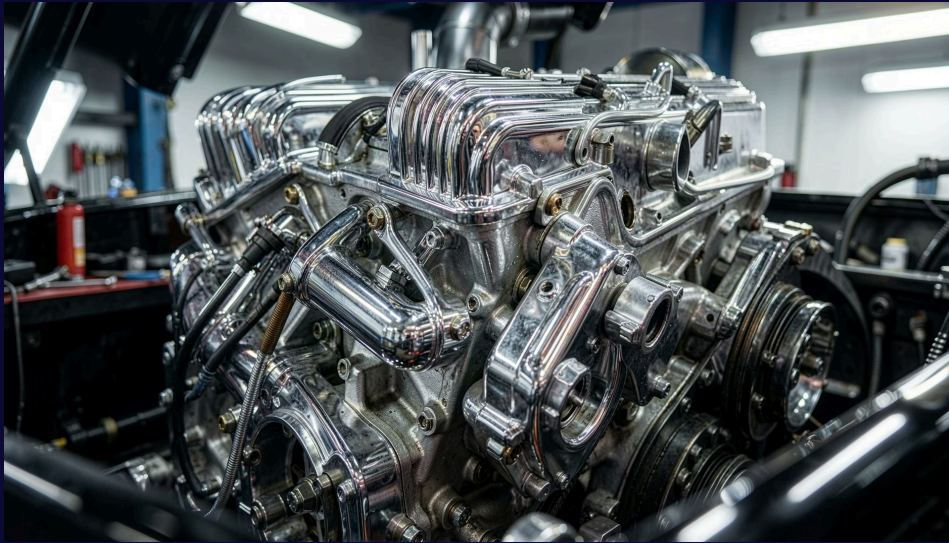
Brain / Logic / Interaction

Teaching Tip: Explain the trio — HTML = Structure, CSS = Design, JavaScript = Brain/Logic. This analogy helps students instantly understand each language's role.



JavaScript Environment

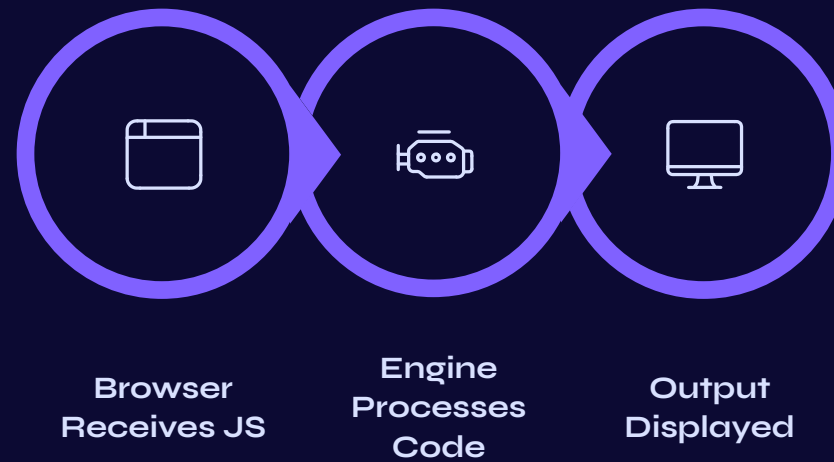
JavaScript runs inside browsers, servers, and mobile apps. Every modern browser contains a built-in JavaScript engine that reads and executes your code instantly.



Real World Analogy

Think of it like a car:

- **Car = Browser**
Chrome, Firefox, Edge
- **Engine = JS Engine**
Chrome uses the V8 Engine
- **Fuel = JS Code**
Your JavaScript powers it all



The browser acts as the container, the JavaScript engine does the heavy lifting, and your code is the fuel that drives everything.

🕒 **Teaching Tip:** Open the browser console and run `console.log("Hello JavaScript");` — students feel excited instantly when they see their first output!

How JavaScript Works

Understanding the execution flow helps students grasp what happens behind the scenes every time a browser runs JavaScript code.



From writing a single line to seeing output on screen, the browser handles everything automatically through its built-in JavaScript engine.

What JavaScript Can Do

 **Change HTML & CSS**
Modify content and styles dynamically

 **Handle Clicks**
Respond to user interactions

 **Fetch Data**
Load content without page reload

Live Example

```
alert("Welcome Students");
```

This single line triggers a popup in the browser.

```
document.body.style.backgroundColor = "black";
```

This changes the entire webpage background color instantly.

 **Teaching Tip:** Change the webpage background color live using JavaScript in the browser console. Students love seeing instant visual results!

Variables in JavaScript

A variable stores data — just like a bottle stores water. Variables are the foundation of every JavaScript program, letting you save and reuse information throughout your code.

Real Life Analogy

 **Bottle** → Stores Water

 **Variable** → Stores Data

Just as a bottle can hold any liquid, a variable can hold any value — a name, a number, or even true/false.

Variable Rules

- Cannot start with a number
- No spaces allowed in names
- Use meaningful, descriptive names

Good vs Bad

```
//
```

let, const and var

JavaScript provides three ways to declare variables. Knowing when to use each one is essential for writing clean, modern JavaScript code.

let — Changing Values

Use when the value needs to change over time.

```
let marks = 80;  
marks = 90; //
```

Data Types in JavaScript

A data type defines what kind of data is stored in a variable. JavaScript has several built-in data types that every developer must know.

Core Data Types

Type	Example
String	"Rahul"
Number	25
Boolean	true
Undefined	undefined
Null	null

Examples in Code

```
let name = "Rahul"; // String
let age = 21;       // Number
let isStudent = true; // Boolean
```

typeof Operator

```
console.log(typeof age);
```

✔ Output: number

Use `typeof` to check what data type a variable holds at any time.

66

String
Text wrapped in quotes

#

Number
Integers and decimals

☐

Boolean
true or false only

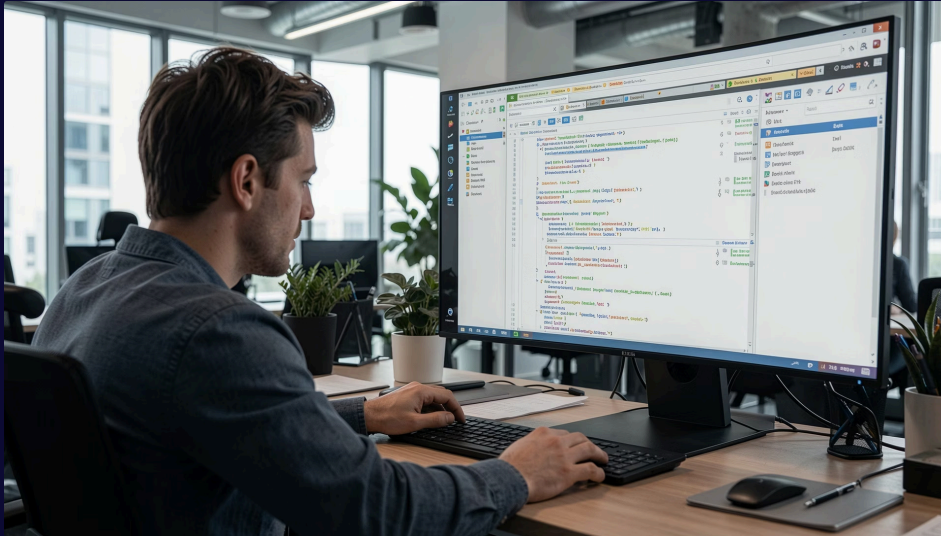
?

Undefined / Null
Empty or unknown values

📌 **Teaching Tip:** Show students a list of values and ask them to identify the data type. This quick quiz reinforces understanding instantly.

Console Methods

The browser console is a developer's best friend. It helps you test code, find errors, and debug your JavaScript programs in real time.



How to Open the Console

01

Right Click

Right-click anywhere on the webpage

02

Inspect

Select "Inspect" from the menu

03

Console Tab

Click the "Console" tab to start coding

1

`console.log()`

```
console.log("Hello");
```

Prints normal output to the console

2

`console.error()`

```
console.error("Error Found");
```

Displays an error message in red

3

`console.warn()`

```
console.warn("Warning");
```

Shows a yellow warning message

Teaching Tip: Run multiple console examples live in class. Let students type each method themselves and observe the different colored outputs.

Mini Practical Task

Build a Student Information Console App

Now it's time to put everything together! Students will create a mini JavaScript program that stores and displays their own personal information using variables and console methods.

Requirements

Each student should store the following using variables:

Name

Store your full name as a String

Age

Store your age as a Number

Course, College & Skills

Store additional details about yourself

Example Code

```
let studentName = "Rahul";
let age = 21;
let course = "BCA";
let college = "Delhi University";
let skills = "HTML, CSS, JS";

console.log(studentName);
console.log(age);
console.log(course);
console.log(college);
console.log(skills);
```

Expected Output

```
Rahul
21
BCA
Delhi University
HTML, CSS, JS
```



Teaching Tip: Ask every student to use their own real information. Personal data makes the exercise more engaging and memorable!

Homework & Final Outcome

After completing Day 1, students will have a solid foundation in JavaScript basics. Here's what to practice at home and what you'll have achieved by the end of today.

Homework Assignment

Create **20 variable examples** practicing all concepts learned today:

- Use `let` for changing values
- Use `const` for fixed values
- Include all data types
- Print each with `console.log()`

```
let city = "Delhi";  
const country = "India";  
let isOnline = true;
```

What You've Achieved Today



Understand JavaScript Basics

What JS is and where it runs



Create Variables

Using `let`, `const`, and data types



Use the Console

`log`, `error`, and `warn` methods



Write Basic JS Programs

Real working JavaScript code

 **Congratulations!** You've taken your first steps into the world of JavaScript. Keep practicing, keep building, and remember — every expert was once a beginner.